Vivasayiai: Tamil Nadu Farming Assistant

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Abstract / Executive Summary

Vivasayiai is an AI-powered farming assistant designed to support farmers in Tamil Nadu by providing easy access to farming-related information. The platform allows farmers to ask questions in Tamil or English and receive guidance related to crops, weather conditions, soil health, and farming best practices through a simple chat interface.

The system uses a modern AI approach that combines a language model with a curated agricultural knowledge base to ensure accurate and relevant responses. Real-time weather and soil data are integrated to provide location-aware recommendations. Vivasayiai is built using a serverless cloud architecture to ensure scalability, reliability, and cost efficiency. The platform aims to improve decision-making for farmers and act as a digital extension support system.

Objectives and Scope

Objectives

- **Localized Farming Guidance:**
Provide farmers with timely and relevant advice based on local weather, soil conditions, and crop requirements.

- **Multilingual Support:**
Enable farmers to interact with the system in Tamil or English for better accessibility.

- **Scalable Digital Assistance:**
Use AI and cloud technologies to support a large number of users without manual intervention.

- **Simple User Experience:**
Design an easy-to-use chat-based interface that works on both desktop and mobile devices.

- **Reliable Information Delivery:**
Ensure responses are generated using verified agricultural knowledge and real-time data sources.

Scope

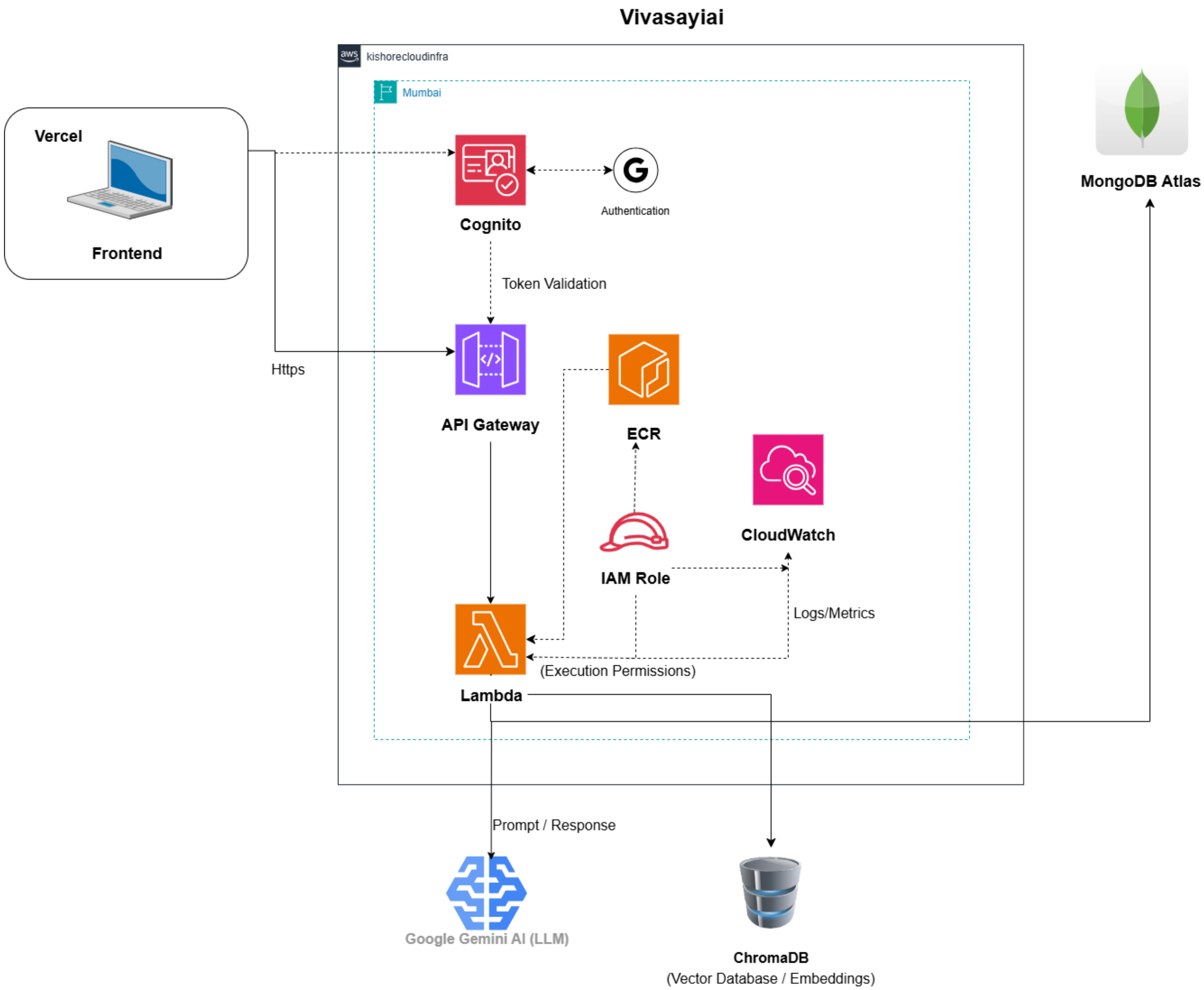
The project includes a farmer-facing chatbot for agricultural queries, backend AI processing using a retrieval-based system, and integration with external weather APIs. The system focuses on advisory support and does not replace agricultural experts.

System Architecture

Vivasayiai follows a cloud-native, serverless architecture. Farmers access the platform through a web browser, which loads the frontend application hosted on Vercel. The frontend handles user interaction, language selection, and chat input.

All requests are sent securely over HTTPS to AWS API Gateway, which routes them to AWS Lambda functions. The Lambda functions handle authentication, query processing, and AI orchestration. The AI logic uses a retrieval-based approach where relevant agricultural documents are fetched from a vector database and combined with a language model to generate accurate responses.

Weather and soil data are fetched from an external weather API based on the user's location. User data and chat history are stored in a database. Logs and metrics are captured using AWS CloudWatch for monitoring and debugging. This architecture ensures scalability, security, and low operational overhead.



Tech Stack Overview

Frontend

- React.js
- Tailwind CSS
- Vite
- Axios
- Google OAuth

Backend

- Node.js
- Express.js
- AWS Lambda
- LangChain
- Vector Database (ChromaDB)

External Services

• Google Gemini (LLM)
• Open-Meteo API (Weather & Soil Data)

Deployment & Monitoring

• Vercel
• AWS API Gateway
• Amazon ECR
• AWS CloudWatch

Features Overview

Farmer-Facing Features

• **User Authentication:**
Google Sign-In for simple and secure access.

• **Language Selection:**
Support for Tamil and English.

• **AI Chat Interface:**
Farmers can ask questions related to crops, weather, irrigation, and farming practices.

• **Weather-Aware Responses:**
Advice includes real-time weather and soil information.

• **Responsive Design:**
Works smoothly on mobile phones and desktops.

Admin / System Features

• Knowledge base management
• Document indexing for AI retrieval
• Monitoring and logging through CloudWatch
• Secure API access control

Folder Structure

Frontend

• components/ – Chat UI components (chat window, messages, sidebar, voice input)
• context/ – Authentication and user session state
• services/ – Backend API and weather service communication
• config/ – Static configuration such as district coordinates
• i18n.ts – Tamil and English language translations

• types.ts – TypeScript interfaces
• config.ts – Environment configuration

Backend

• routes/ – Chat, authentication, and weather API routes
• controllers/ – Request handling and business logic
• services/ – AI (RAG), chat context, and weather services
• models/ – Chat session and message schemas
• utils/ – Helper and utility functions
• config/ – Database and environment configuration
• deploy-lambda.yml – CI/CD workflow for AWS Lambda deployment

Key API Endpoints

Authentication

• POST /api/auth/google – Google Sign-In authentication

Chat & AI

• POST /api/chat – Send user message with context
• GET /api/chat/session/:chatId – Fetch a specific chat session
• GET /api/chat/sessions – Fetch all user chat sessions

Deployment Overview

• Frontend: Vercel
• Backend: AWS Lambda with API Gateway
• AI Model: Google Gemini
• Vector Store: ChromaDB
• Monitoring: AWS CloudWatch
• CI/CD: GitHub Actions

This deployment ensures automatic scaling and cost-efficient operation.

Implementation & Contributions

As the sole developer of Vivasayiai, I:

• Designed the system architecture
• Built the frontend using React and Tailwind CSS
• Developed backend APIs using Node.js and Express
• Implemented the RAG-based AI workflow

• Integrated Google OAuth authentication
• Connected weather and soil data APIs
• Deployed the system using AWS Lambda and Vercel
• Set up CI/CD pipelines and monitoring

Challenges & Solutions

• AI Accuracy:
Used a retrieval-based approach to reduce incorrect responses.

• Multi-language Handling:
Ensured proper input and output handling for Tamil text.

• Serverless Deployment:
Optimized Lambda performance and cold-start handling.

• API Integration:
Handled rate limits and failures with retries and fallbacks.

Future Enhancements

• Voice-based interaction for farmers
• Market price integration
• SMS and WhatsApp support
• Offline access for low-connectivity areas
• Advanced personalization using user feedback

Conclusion

Vivasayiai is a scalable and practical AI-powered farming assistant built using modern cloud and AI technologies. By combining local language support, real-time weather data, and intelligent information retrieval, the platform helps farmers make informed decisions. The serverless architecture ensures reliability and cost efficiency, making Vivasayiai a strong foundation for future digital agriculture solutions.

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